Parallelism and Blockchain

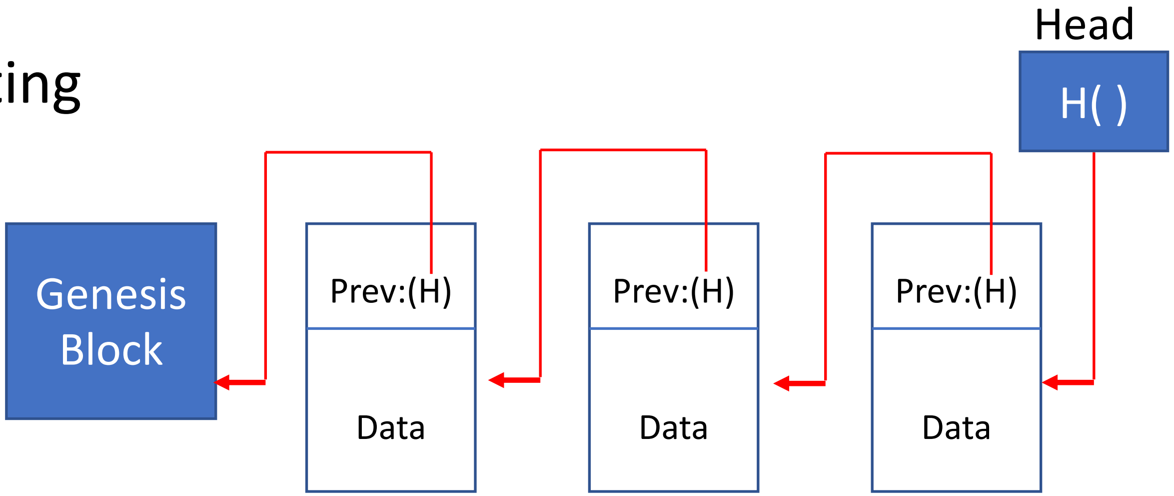
CSCI 654 Foundations of Parallel Computing

Blockchain

- Blockchain
  - Immutable decentralized database
  - Transparent
  - Auditable
  - Started as the underlying technology for Bitcoin (2009)
    - Problems with scale

What is a Blockchain ?

- Distributed ledger
  - Blocks of consensus data representing transactions
    - Synchronized
    - Distributed
    - Replicated
- A hash chain
  - Each block verifies integrity of the entire blockchain preceding it
    - Linked list
      - Cryptographic hash pointers to previous block



- Tampering with previous data is detectable
- Transactions are validated
- Blockchain can be extended

Blockchain scalability

- Transaction throughput
  - Rate at which transactions are processed
  - Poor compared to traditional transactions
    - Couple of magnitudes lower
- Latency
  - Time to confirm inclusion of a transaction in the blockchain
    - Couple of magnitudes longer

What is a transaction?

- A transformation of the state of the ledger
  - A transaction is subjected to validity and verification checks
    - Payers and payees are identified by their public keys
    - The payer digitally signs a transaction
    - Nodes in the network verify
      - Integrity of the transaction
      - Payer is authorized to make the transaction
      - Sum of outputs is less than the sum of inputs
      - No double spending
  - Transactions are included in a block of the blockchain
  - A leader coordinates with nodes to reach consensus and appends a final committed block to the chain
    - Honest leader

Leader or miner

- Only leader nodes can append information to the blockchain
  - Leader is chosen randomly via proof of work
    - Solving a hash puzzle – highly computation intensive

Consensus mechanisms – choosing miner

- Proof of work
  - Requires high computational power
- Proof of stake
  - Miner is chosen based upon proportional stake
- Proof of capacity
  - Requires hard disk capacity for mining rather than computing power
- Proof of burns
  - In crypto currency – based on amount of coins burned
- Proof of activity
  - Combination of work and stake
- Proof of elapsed time
  - Sleep and wakeup

Parallel

- Multiple leaders
  - Rather than choosing one leader at each epoch
  - Choose multiple leaders
    - Multiple leaders collectively decide the next block to be appended
    - Combined parallel work of leaders
    - Leader group – communication network
    - PBFT to generate a collective signature* --- Practical Byzantine Fault Tolerance protocol

Parallel Blockchain

- Traditional blockchain is a linear process
- Multiple leaders can extend in different parts of the blockchain
- Graphs instead of blocks and chains
  - A transaction is validated by multiple children
  - Miners extend different branches of the transactions graph in parallel

Sharding Transactions

- Nodes are grouped into committees
  - Hierarchical
  - Each committee manages a subset of transactions (shards)
  - Within a committee, nodes use PBFT to agree on a block of transactions
  - Committee upon agreement, sends the block to a final committee
  - Transaction throughput increases with number of nodes